

Full Stack Development with MERN

Project Documentation format

1. Introduction

- Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- Team Members: Dharshini M — Individual project member / administrator.
- Platform: ServiceNow.

2. Project Overview

- Purpose: Automate the standard laptop request-to-fulfillment handoff so that a requester can submit a standardized laptop request and ServiceNow can automatically create the fulfillment task.
- Features: Standard Laptop Service Catalog item; Service Catalog trigger; Flow Designer / Workflow Studio automation; Create Catalog Task action; request, RITM and SCTASK traceability.

3. Architecture

- Frontend / User Interface: ServiceNow Service Catalog for the requester and ServiceNow platform interface for administrators.
- Backend / Logic: Flow Designer / Workflow Studio manages the automation. No separate Node.js or Express.js backend is used.
- Database: ServiceNow-managed request, Requested Item (RITM), and Catalog Task records. No separate MongoDB database is used.

4. Setup Instructions

- Prerequisites: Access to the ServiceNow instance with appropriate roles/permissions.
- Configuration: Create the Standard Laptop catalog item under Hardware, configure the Process Engine / Service Catalog mapping, configure the flow trigger, and add the Create Catalog Task action.
- Validation: Submit a Standard Laptop request and verify the generated request, RITM and Catalog Task records.

5. Folder Structure

- Configuration Structure: Service Catalog Item → Process Engine / Flow → Service Catalog Trigger → Create Catalog Task → Catalog Task.
- Unlike a MERN application, this project does not use separate client/server folders because the implementation is configured directly within ServiceNow.

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6. Running the Application

- Open the ServiceNow Service Catalog.
- Select the Standard Laptop catalog item and submit the request.
- The Service Catalog trigger starts the configured Flow Designer / Workflow Studio flow.
- The Create Catalog Task action generates the IT fulfillment task linked to the request.

7. API Documentation

- No external API endpoints are required for the core solution.
- The project uses native ServiceNow platform capabilities and records rather than a separate REST API backend.

8. Authentication

- Authentication is handled by the ServiceNow instance.
- Roles and access controls / ACLs determine which users can submit requests, configure flows, or access fulfillment records.

9. User Interface

- Requester interface: ServiceNow Service Catalog where the Standard Laptop item is selected and submitted.
- Administrator interface: ServiceNow platform and Flow Designer / Workflow Studio for configuration and monitoring.

10. Testing

- UAT coverage includes Standard Laptop item availability, Service Catalog submission, flow triggering, Catalog Task creation, request/RITM records and request-to-task traceability.
- Documented evidence includes REQ0010002, RITM0010002 and SCTASK0010001.
- Known test observations: Assigned-to value was not confirmed in the available task evidence, and no quantitative load/performance benchmark was captured.

11. Screenshots or Demo

- Demonstration sequence: Standard Laptop catalog item → Process Engine / Service Catalog mapping → Flow Designer trigger → Create Catalog Task → REQ0010002 → RITM0010002 → SCTASK0010001.
- Use the supplied ServiceNow screenshots/evidence for the catalog item, flow configuration, request, RITM and Catalog Task.

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12. Known Issues

- The Assigned-to value is not confirmed in the supplied SCTASK evidence.
- A quantitative load/performance benchmark was not captured during the available testing.

13. Future Enhancements

- Add assignment rules to automatically route hardware tasks to the appropriate fulfillment group or user.
- Add approval steps for laptop procurement where institutional policy requires approval.
- Add automated requester notifications for request, task and completion status changes.
- Configure SLAs and escalation rules for fulfillment tasks.
- Extend the same catalog-and-flow pattern to other standard IT hardware and services.
- Add procurement analytics and reporting for request volumes, fulfillment time and service performance.

Project Summary

The solution standardizes laptop procurement through the ServiceNow Service Catalog and automates the handoff to IT fulfillment using Flow Designer / Workflow Studio. The core implementation avoids a separate application backend and keeps the request, RITM and Catalog Task within the ServiceNow platform.